Nguyen Van Cuong

Junior AI Engineer

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OBJECTIVE

- AI Engineer with strong passion for technology and 3 years of experience in system development (Web Development). Currently focusing on advanced research in Deep Learning, Computer Vision, and Natural Language Processing (NLP). Eager to apply artificial intelligence to solve real-world problems and create value for the community. Seeking opportunities in a high-tech environment that encourages collaboration and problem-solving to help organizations achieve their AI strategic goals.

SKILLS

Languages	Python, JavaScript, TypeScript
AI & Machine Learning	AGNO, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM
Generative AI & LLM	HuggingFace, LangChain, LlamaIndex, RAG, AI Agents
Computer Vision	OpenCV, YOLO (v8/v11), Object Detection, Image Classification
Data Engineering	Pandas, NumPy, Feature Engineering, Data Visualization
Web Development	React.js, Next.js, Strapi CMS, RESTful API

WORK EXPERIENCE

Derasoft Co., Ltd 2021 - 2026

Web Developer

- Collaborated with clients to understand requirements, solve technical issues, and deliver effective web solutions.
- Developed and maintained websites using ReactJS, NextJS, and Strapi CMS, including Headless CMS integration and CI/CD deployment.
- Optimized website performance, SEO, and UX/UI while supporting data-driven improvements with cross-functional teams.

PROJECTS

AI Blog-to-Podcast Converter | Streamlit, OpenAI, Firecrawl, ElevenLabs 2025 - 2026

Personal Project

- Developed an end-to-end application that automates the transformation of web-based blog content into engaging audio podcasts.
- **Content Extraction:** Integrated **Firecrawl API** to scrape and clean full-text data from diverse blog structures.---
- **AI Summarization:** Leveraged **GPT-4** to synthesize complex articles into concise, engaging scripts (under 2,000 characters).
- **Neural TTS Integration:** Utilized **ElevenLabs API** for high-quality, realistic voice synthesis to generate final audio episodes.
- **User Interface:** Designed an intuitive dashboard using **Streamlit** with secure API key management for personalized user configurations.

AI Travel Agent | Streamlit, OpenAI GPT-4o, Python-ICS 2025 - 2026

Personal Project

- Developed an intelligent AI-powered travel agent that automates complex vacation research and itinerary organization.
- **Intelligent Planning:** Leveraged **GPT-4o** to generate highly personalized travel plans, including destination discovery, activity curation, and accommodation recommendations.
- **Dynamic Customization:** Implemented logic to tailor itineraries based on user-defined trip durations and specific travel preferences.
- **Calendar Integration:** Developed a feature to convert AI-generated plans into **.ics files**, enabling seamless synchronization with Google Calendar and other calendar applications.
- **User Experience:** Designed a responsive and intuitive dashboard using **Streamlit** to streamline the vacation planning workflow.

Agentic RAG Agent (Local Execution) | Ollama, Llama 3.2, LanceDB, Streamlit 2025 - 2026

Personal Project

- Engineered an agentic Retrieval-Augmented Generation (RAG) system to enable intelligent querying and interaction with PDF-based knowledge bases.
- **Local LLM Integration:** Leveraged **Llama 3.2** and **EmbeddingGemma** via **Ollama** to ensure data privacy and efficient local inference.
- **Advanced Vector Search:** Implemented **LanceDB** for high-performance vector storage and semantic search, enabling dynamic document ingestion from PDF URLs.
- **Agentic Workflow:** Developed a tool-calling logic that allows the model to autonomously decide when to retrieve information or synthesize answers.
- **Real-time Interface:** Designed a **Streamlit** dashboard featuring streaming responses and "thought-process" visibility for enhanced user interaction.

Agentic RAG Knowledge Assistant | Agno, GPT-5, LanceDB, Streamlit 2025 - 2026

Personal Project

- Engineered an advanced Agentic Retrieval-Augmented Generation system designed for intelligent knowledge retrieval and sophisticated decision-making.
- **Agentic Orchestration:** Leveraged the **Agno framework** to build autonomous agents capable of intelligent tool-calling and multi-source knowledge synthesis.
- **State-of-the-Art Inference:** Integrated **GPT-5** to handle complex reasoning tasks, providing highly accurate and context-aware responses.
- **High-Performance Retrieval:** Implemented **LanceDB** for efficient vector storage and rapid similarity search, enabling dynamic expansion of the knowledge base via URLs.
- **Real-time UX:** Developed a streamlined **Streamlit** interface featuring live response streaming and structured Markdown output for enhanced readability.

Real-time Poker Hand Recognition System | YOLOv8, OpenCV, PyTorch 2025 - 2026

Personal Project

- Developed an end-to-end computer vision application to detect 52 classes of Western playing cards and evaluate Poker hand rankings in real-time.
- **Object Detection:** Fine-tuned and deployed a **YOLOv8** model to achieve high-accuracy detection of card suits and ranks from live webcam streams.
- **Logic Implementation:** Engineered a robust evaluation algorithm to normalize detected data and classify Poker hand types (e.g., Royal Flush, Straight, Flush) based on a 5-card input.
- **Stream Optimization:** Integrated **OpenCV** for efficient frame processing, ensuring real-time performance for bounding box visualization and result overlay.

Vietnamese Emotion Recognition System | Python, Scikit-learn, SVM, NLP 2025 - 2026

Personal Project

- Engineered a high-performance sentiment analysis system to classify Vietnamese text into 7 emotional states (Anger, Disgust, Enjoyment, Fear, Other, Sadness, Surprise) with **92.7% accuracy**.

- **Advanced NLP Pipeline:** Architected a comprehensive preprocessing workflow tailored for Vietnamese, including abbreviation normalization, profanity filtering, and emoji-to-emotion token mapping. Utilized underthesea for precise Vietnamese word segmentation.
- **Data Optimization:** Addressed class imbalance issues by implementing **Oversampling** techniques, ensuring the model's robustness across all emotional categories.
- **Feature Engineering & Modeling:** Leveraged **TF-IDF Vectorization** with N-gram (1,2) features to capture local context. Trained a **Linear SVM** model, optimized for high-dimensional sparse text data.
- **Evaluation:** Achieved a superior **F1-macro score of 0.927**, demonstrating the system's ability to handle complex linguistic nuances and "noisy" social media text.

ALL Project 2025 - 2026

Personal Project

- For more project details and live demos, please visit: https://github.com/ncuong0704/About-me

EDUCATION

Ho Chi Minh City University of Industry and Trade (HUIT) 2013 - 2016

Bachelor of Business Administration

- Focused on: *Statistics, Excel-based Data Analysis, Market Research, and Business Decision-Making.*

CERTIFICATIONS

- **Top 1 - Applied Machine Intelligence Specialist** – Cybersoft 2025
- **Top 5 - Fundamentals of Data Analyst and AI Automation** – Cybersoft 2025
- **Data Science/Machine Learning/Python Advanced** – Viet Nguyen AI 2025
- **Deep Learning for Computer Vision (Basic & Advanced)** – Viet Nguyen AI 2025
- **Certificate of Proficiency in Information System Management (CPISM)** - *Aptech* 2022

COURSES IN PROGRESS

- **Advanced AI Engineering with NLP, LLM, RAG, RAGFlow, LangChain, Generative AI, and AI Agents** - *Cypersoft* In progress
- **ML Ops/Cloud – AI Deployment** - *Cypersoft* In progress

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